

A Study and Evaluation of Graph RAG on Wikipedia Data

Vietnamese Multi-hop Question Answering over Wikipedia

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Research Problem and Objectives

Problem

Vietnamese multi-hop QA needs complete, traceable evidence across Wikipedia pages. Text-only retrieval can miss the relation connecting the required facts.

- 1 Build a reproducible Vietnamese Wikipedia corpus with stable evidence IDs.
- 2 Combine lexical, dense, graph, and community retrieval in Neo4j.
- 3 Evaluate answer, supporting evidence, abstention, latency, and ablations.

Users, Applicability, and Innovation

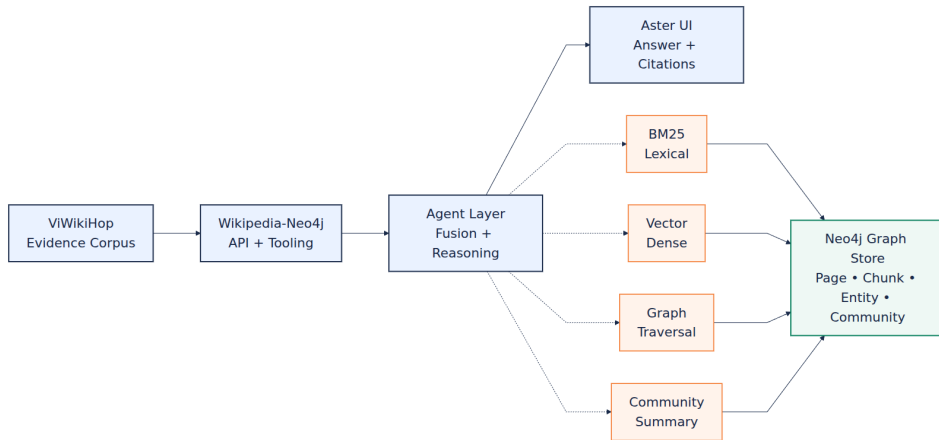
Real users and value

- Students and researchers need faster evidence synthesis across multiple Wikipedia pages.
- Journalists and analysts need verifiable Vietnamese citations rather than answer-only summaries.
- AI developers can reuse the corpus, graph schema, and evaluation pipeline.

What is new in our system

- Local-first Vietnamese GraphRAG with auditable evidence IDs.
- WRRF combines lexical, vector, graph, and community retrieval signals.
- ViWikiHop adds answerable and unanswerable multi-hop items with quality control.

System Architecture



ViWikiHop controls corpus evidence, Wikipedia-Neo4j performs retrieval and reasoning, and Aster presents the live QA

Dataset Motivation

Dataset	Size	Role	Stage
Vietnamese Wikipedia (final corpus)	~590K pages	Corpus and graph construction	R1–R3
UIT-ViQuAD 2.0	36,457 QA pairs	Retrieval benchmark	R1–R2
ViWikiHop	1,600 QA items	Multi-hop end-to-end evaluation	R2–R3

- Review 1 focused on feasibility: identify real Vietnamese sources and measurable benchmarks.
- Review 2 and Review 3 converted that plan into a grouped, answerable/unanswerable test set for GraphRAG evaluation.

Data Pipeline — Corpus Filtering

11-Stage Filter: viwiki-20260501 Snapshot

Stage	Count	Drop
Initial dump	1,614,015	—
Redirect filter	1,300,130	−313K
Disambiguation	1,284,671	−15K
List pages	1,279,292	−5K
Year/date pages	1,276,304	−3K
Dup-line filter	1,275,171	−1K
Word count pre	414,397	−861K
Orphan filter	397,438	−17K
Word count final	357,747	−40K
Stop words	357,736	−11
Exact dedup	357,714	−22
MinHash dedup	353,498	−4K

Key Design Choices

- Word count pre-filter: largest single drop (−861K stubs)
- Orphan filter: removes unlinked articles (−17K)
- MinHash dedup: Jaccard ≥ 0.85 threshold
- **No** punctuation-ratio filter: Wikipedia tables/lists naturally break the 60% threshold; dropped after analysis

Result

1,614,015 → **353,498** articles
 (21.9% of original dump retained)
 Snapshot: viwiki-20260501

This slide reports the Review 2 filtering experiment snapshot; later production ingestion uses a different cleaned corpus snapshot.

Corpus Construction

Final production corpus used in the current system

Stage	Retained pages	Removed	Retention
Input snapshot	1,288,472	–	100.00%
Structural filters	884,219	404,253	68.63%
Quality filters	641,738	242,481	49.81%
Orphan filter	598,314	43,424	46.44%
Deduplication	589,742	8,572	45.77%

- Removes redirects, lists, weak Vietnamese pages, duplicates, and graph orphans.
- Retains cleaned text, wikilinks, provenance, and stable article-local segment IDs for downstream ingestion.

Data Pipeline — ViWikiHop Schema v0.4

Entry Schema (schema v0.4)

Field	Description
_id	hw- / syn-mh- / uvq-
question	Vietnamese question
answer	Empty if unanswerable
supporting_facts	Evidence refs
context	Paragraph segments
type	8 reasoning types
level	easy / medium / hard
source	3 sources
num_hops	≥ 1
is_vi_exclusive	VI-exclusive track
is_impossible	Unanswerable flag
decomposition	Sub-Q/A per hop
plausible_answer	For unanswerable only

3 Data Sources

Source	Description
hw-	Human annotated
syn-mh-	KG walk synthesis
uvq-	ViQuAD 2.0 import

Schema v0.4 Key Invariants

- Single-hop: `decomposition = []`
- Unanswerable: `answer=""`, non-empty `plausible_answer`
- Multi-hop: `|decomposition| = num_hops`
- UIT-ViQuAD: `is_vi_exclusive` always false

Data Pipeline — Reasoning Types & Difficulty

8 Reasoning Types

Type	Description
lookup	One article, one fact
bridge	A $\rightarrow$ bridge entity $\rightarrow$ B
comparison	Compare ≥ 2 article values
intersection	Entity from ≥ 2 constraints
temporal	Date order / chronology
numerical	Arithmetic or counting
fan-out	Aggregate related entities

3-D Difficulty Rubric

Score = hop_count + lexical_overlap + reasoning_op

Score	Hop	Lexical	ReasonOp
0	1-hop	$\geq 60\%$	lookup
1	2-hop	30–60%	bridge/comparison
2	≥ 3 -hop	$< 30\%$	temporal/numerical/...

Total (0–6)	Level
0–1	easy
2–3	medium
4–6	hard

Data Pipeline — QA Generation & QC

QA Generation Pipeline

- ➊ **Bridge Extraction** — follow `[[wikilinks]]` to build article-pair bridges; no Neo4j walk
- ➋ **LLM Sampling** — prompt LLM to generate questions directly from bridge pairs (no templates)
- ➌ **Labelling** — **Gemini 2.5 Flash Lite** reads difficulty rubric and assigns type + level
- ➍ **Verification** — **DeepSeek V3 Pro** independently verifies label and answer grounding

Sample Entry ID

```
syn-mh-viex-b1-a-00001
type=bridge, level=medium, num_hops=2
is_vi_exclusive=true, source=synthetic-multihop
```

Two-Model QC Gate

Model	Role
Gemini 2.5 Flash Lite	Label (type, level, decomposition)
DeepSeek V3 Pro	Verify label + grounding

Gate logic: entry passes only when both models agree on answer grounding; label is taken from Gemini, rejected if DeepSeek flags a mismatch.

Schema QC (post-gate)

- Well-formedness: non-empty fields, schema v0.4
- Grounding: answer in `supporting_facts` verbatim
- Natural phrasing: no “*Theo bài viết*” phrases

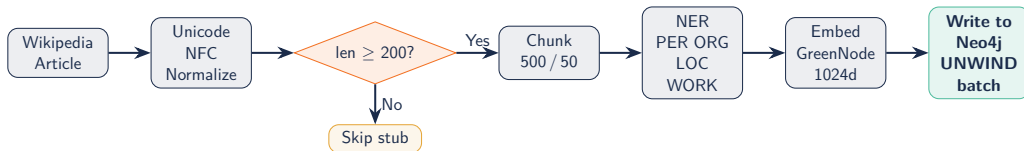
ViWikiHop Dataset and Quality Controls

Split	Answerable	Unanswerable	Total
Train	783	337	1,120
Development	168	72	240
Test	167	73	240
Total	1,118	482	1,600

Quality controls

- Group-aware split to reduce leakage across related questions.
- Includes both answerable and unanswerable items for abstention testing.
- Manual review and schema checks after automatic generation.

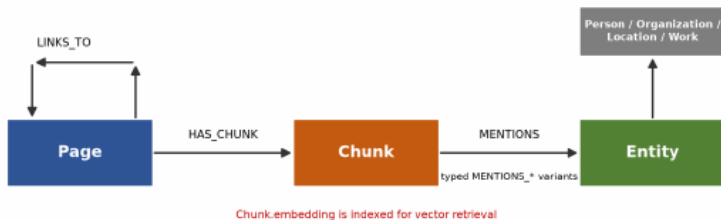
AI Pipeline — Data Ingestion



Stub filter	NER backend	Embedding
1.29M → ~590K cleaned articles	wikilink (Typed F1 = 46.9%)	GreenNode 1024-dim

Knowledge Graph and Retrieval

Core Neo4j property-graph schema



- Nodes: Page, Chunk, typed Entity, and Community; edges preserve mentions and article links.
- Four retrieval channels: BM25, dense vectors, entity graph, and community summaries.

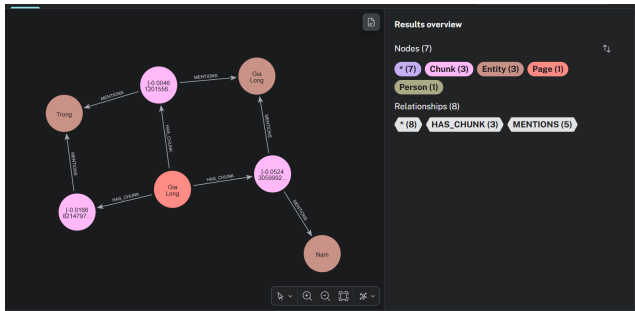
Neo4j Ingestion Status

Retrieval-ready graph

Metric	Value	Notes
Pages	2,339	articles indexed in current Neo4j snapshot
Chunks	31,470	segmented evidence units in that snapshot
Entities	22,736	total entity nodes
MENTIONS	136,080	chunk-to-entity links
LINKS_TO	4,956	page-to-page links
Communities	46	clustered summaries
Chunk embeddings	31,470 / 31,470	100% coverage
Typed entities	20,560 / 22,736	~90.4% typed coverage
Generic entities remaining	2,241	untyped tail after extraction

This graph is the retrieval snapshot used for the reported experiments, not the full cleaned Wikipedia corpus.

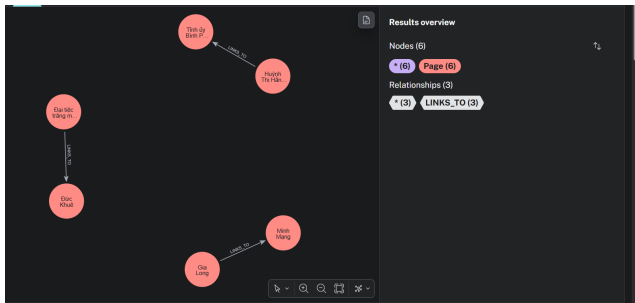
Neo4j Backbone — Page to Chunk



Graph backbone

- Page - `[ :HAS_CHUNK ] -> Chunk`
- Pages preserve article-level provenance while chunks keep retrieval at evidence granularity.
- This backbone ties every evidence unit to its source document.

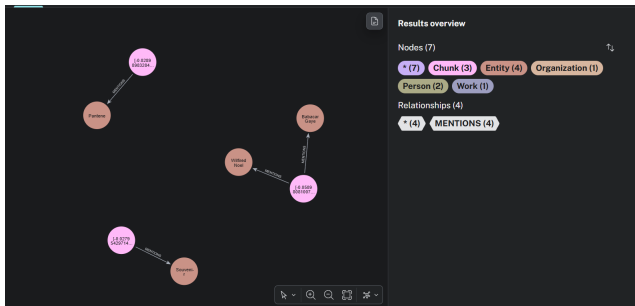
Neo4j Backbone — Chunk to Entity



Document and entity layer

- Page -[:LINKS_TO]-> Page
- Typed entities include Person, Organization, and Work.
- This layer preserves article navigation and semantic typing together.

Neo4j Backbone — Page Links and Typed Entities



Mention structure

- Chunk -[:MENTIONS]-> Entity
- Entity nodes support graph traversal beyond raw text retrieval.
- Mentions connect evidence chunks to reusable semantic nodes.

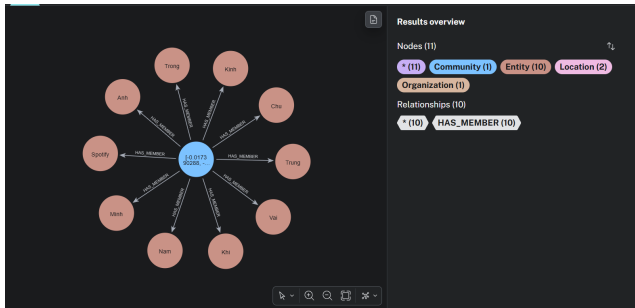
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- Entity
-[:BORN_IN|MEMBER_OF]->
Entity
- Higher-level relations support multi-hop evidence chains.
- Retrieval can move from mentions to semantic relations when text alone is insufficient.

Neo4j Semantic Relations — Community Layer



Community layer

- Community -[:HAS_MEMBER]-> Entity
- Communities summarize clusters of related entities and pages.
- This layer helps the retrieval system surface broader thematic evidence.

Algorithm Complexity and Design Rationale

Technique	Role	Why it remained in the final system
WRRF	Rank fusion	Robustly combines heterogeneous retrieval channels
Graph traversal	Evidence chaining	Improves multi-hop support coverage
Cross-encoder reranking	Candidate ordering	Lifts MRR after fusion
Agent routing	Question handling	Separates simple and complex reasoning paths

Weighted Reciprocal Rank Fusion

Fusion score

$$S_{\text{WRRF}}(d) = \sum_{i \in \{b, v, g, c\}} \mathbb{I}[d \in L_i] \frac{w_i}{k + r_i(d)}, \quad k = 60.$$

- Fuses rankings rather than incompatible raw scores.
- Deduplicates candidates by `chunk_id`.
- Cross-encoder reranks the fused candidates.
- Link expansion supports multi-hop evidence chains.

Evaluation Protocol

Three settings

- 1 Gold-context QA
- 2 Open-corpus retrieval
- 3 End-to-end QA

Metrics

Hit@5, MRR, support recall, complete-chain recall, EM, F1, joint F1, abstention accuracy, and latency.

- Same ordered 240-item test set, graph snapshot, models, and seeds 42, 43, and 44.

Final Retrieval Results

Method	Hit@5	MRR	Support recall	Chain recall	Latency
BM25	0.552	0.351	0.438	0.241	28.4 ms
Vector	0.601	0.402	0.481	0.276	36.9 ms
Entity graph	0.574	0.384	0.522	0.318	44.7 ms
WRRF	0.684	0.472	0.603	0.391	63.8 ms
WRRF + reranking	0.711	0.501	0.628	0.417	118.6 ms

Fusion provides the

strongest evidence coverage; reranking improves ordering at a measurable latency cost.

Ablation Study

Mode	Hit@5	Δ Hit@5	MRR	Mean latency
Full hybrid	0.711	–	0.501	118.6 ms
Text only	0.552	-0.159	0.351	28.4 ms
Graph only	0.574	-0.137	0.384	44.7 ms
No reranking	0.684	-0.027	0.472	63.8 ms
No multi-hop	0.648	-0.063	0.443	92.1 ms

- Link expansion contributes 0.063 Hit@5.
- Reranking contributes 0.029 MRR while adding 54.8 ms.

End-to-End QA and Answerability

Group	Answer EM	Answer F1	Support EM	Support F1	Joint F1	Abstention	The
All	0.592	0.683	0.529	0.667	0.602	0.792	
Answerable	0.521	0.653	0.443	0.641	0.536	–	
Unanswerable	–	–	0.726	0.726	–	0.753	

remaining gap between retrieval and joint F1 motivates evidence-aware answer generation and abstention.

Error Analysis and Limitations

Observed failure sources

- Incomplete evidence chains
- Entity-resolution noise
- Unsupported answer synthesis
- Incorrect abstention

System boundaries

- Segment-to-chunk alignment remains incomplete.
- NER noise can distort graph neighborhoods.
- Aster chat is live; some visual fixtures are static.

Contributions

- ① A reproducible Vietnamese Wikipedia corpus pipeline with stable, verifiable evidence identities.
- ② ViWikiHop: 1,600 answerable and unanswerable multi-hop QA records with group-aware splits.
- ③ Neo4j GraphRAG combining lexical, vector, graph, and community retrieval through WRRF.
- ④ Support-aware evaluation showing the benefit and latency trade-off of reranking and multi-hop expansion.

Conclusion

Main result

The full GraphRAG configuration achieves **0.711 Hit@5**, **0.501 MRR**, and **0.417 complete-chain recall** on the ViWikiHop test set.

- Hybrid fusion is stronger than individual retrieval channels.
- Link expansion and reranking improve quality, with explicit latency trade-offs.
- Canonical evidence makes multi-hop results inspectable rather than answer-only.

Thank You

Questions & Answers